1. The undirected Hamiltonian Cycle Problem is as follows: given an undirected graph G, is there a cycle which contains *all* the vertices ? The undirected Hamiltonian Path problem is as follows: given an undirected graph G, is there a path which contains all the vertices ? Show that the undirected Hamiltonian path is polynomial time reducible to the undirected Hamiltonian Cycle problem.

Contruction

For an instance $G=(V,E)$ of Hamilatonian Path

Construct a new $G'=(V',E')$ as -

$V' = V \cup \{v_{new}\}$

$E' = E \cup \{(u, v_{new}) \mid u \in V\}$

It takes $O(|V|)$ time

$G \in$ Path iff $G' \in$ Cycle

($\Rightarrow$) Forward : If $G \in$ Path then $G' \in$ Cycle

$\exists$ a path P as $P = (v_1, v_2, \ldots, v_n)$ s.t. $n = |V|$

In our G' construction : v_{new} is connected to $\forall v \in V$

so $\exists$ an edge connecting v_{new} and v_1

v_{new} and v_n

we can construct a cycle as $(v_{new}, v_1, v_2, \ldots, v_n, v_{new})$

Since P already covered all

$P + v_{new}$ is a cycle & covers all $\Rightarrow G' \sim$ h. cycle

($\Leftarrow$) If G' has cycle then G has path

Assume G' had a cycle C'

C' visits every vertex exactly once. Hence v_{new} is visited one

Since v_{new} is in a cycle

$\exists$ two edges as (x, v_{new}) and (v_{new}, y)

If we remove v_{new}

cycle becomes a path from $x \to y$

C' visits all, so the path visits all vertex in $V'/\{v_{new}\}$

Since the path visits all in $V = V'/\{v_{new}\}$,

it is an hamilitanion path ■

2. The Path Selection Problem can be defined as follows: given a directed graph G, a set of directed paths $P_1, \cdots P_r$ in the graph G, and a number k, is it possible to select at least k of these paths so that no two of the selected paths share any vertices? Prove that the Independent Set problem is polynomial time reducible to the Path Selection Problem.

Given instance of IS as $G=(V,E)$ and integer k

Construction

A new $G'=(V',E')$

Create an edge node for all edge
vertex node for all vertex

$V' = \{n_v \mid v \in V\} \cup \{x_e \mid e \in E\}$

Paths:

One directed path for all nodes s.t.
if $e_{v1}, e_{v2}, \ldots e_{vm}$ are incident on v

$P_v : n_v \to x_{v1} \to x_{v2} \to \ldots \to x_{vm}$

Edges:

As per the paths

K':

$K' = k$

Time of construction:

no of nodes in $G' = |V| + |E|$
edges in $G' = \sum d(v) = |E|$

runs in $O(|V|+|E|)$

G has independent Set of size k iff
G' allows selection of K' disjoint paths

If G has independent set of size k, then G' has - - - -

→ Let G has set $S \subseteq V$ s.t. $|S| = k$

For each vertex v in S ∃ a path P_v in G'

TS: $P_1 \cap P_2 \cap \ldots P_k = \phi$

$\forall u, v \in S$. ∄ any edge $E = (u, v)$ in G

P_u and P_v share vertex if they have common edge-node x_e

Due to absence of e_{uv}, paths P_u and P_v do not intersect

n_u and n_v are unique

Hence we got k disjoint paths

If G' has k disjoint paths, G has k independent vertices

let the paths be as

$P = \{P_{u_1}, P_{u_2}, \ldots, P_{u_k}\}$

Corresponding vertices set $S = \{u_1, u_2, \ldots, u_k\}$

let S~ not independent, ∃ $e = (u, v) \in E$

Thus $x_e \in P_u$ & $x_e \in P_v$

This contradicts the fact paths selected P_u & P_v are disjoint

∴ Our assumption was wrong,

all vertices in S are independent.

S~ independent set of $|S| = k$.

3. The Subset-Sum Problem is defined as follows - Given a set of n positive integers $S = \{a_1, a_2, \ldots, a_n\}$ and a target value W, the Subset Sum problem asks whether there exists a subset $T \subseteq S$ such that $\sum_{a_i \in T} a_i = W$.

4. Given a multiset of n positive integers $S = \{a_1, a_2, \ldots, a_n\}$, the 2-Partition problem asks whether S can be partitioned into two subsets S_1 and S_2 such that $S_1 \cup S_2 = S$, $S_1 \cap S_2 = \emptyset$, and $\sum_{a_i \in S_1} a_i = \sum_{a_j \in S_2} a_j$.
Show that SUBSET-SUM is polynomial time reducible to 2-PARTITION.

Construction

let $K := \sum a_i$

W = target sum

Construct S' as $S = S \cup \{x, y\}$

$x := 2K - W$

$y := K + W$

Time: $O(n)$

$SUM(S') = 4K$

$\sum S_1' = \sum S_2' = 2K$

($\Rightarrow$) If S has subset sum = K, S' has an partition

$\exists$ a subset T

$SUM(T) = W$

Remaining elements T'

$SUM(T') = K - W$

$S_1' = T \cup \{x\}$ $\quad S_2' = T' \cup \{y\}$

$S_1' \cup S_2' = S' = T \cup T' \cup \{x, y\} = S \cup \{x, y\}$

$SUM(S_1) = SUM(S_2) = \cancel{y} + 2K - \cancel{y} = K - \cancel{y} + K + \cancel{y} = 2K$

$\Leftarrow$ S' can be partitioned, S has subset sum = W

let $S' = S_1' \cup S_2'$

$SUM(S_1') = SUM(S_2') = 2K$

Can x, y be in same partition? No!

cause: $2K - W + K - W = 3K > 2K$

S_1' contains x

let $T = S_1' / \{x\}$

$SUM(S_1') = 2K \Rightarrow SUM(T) + 2K - W = 2K$

$\Rightarrow SUM(T) = W$

Hence Proved!

5. Define The Zero Weight Cycle problem as follows: you are given a directed graph $G = (V, E)$, with weights w_e on the edges $e \in E$. The weights can be negative or positive integers (or equal to 0). You need to decide if there is a cycle in G so that the sum of the edge weights on this cycle is exactly 0 (i.e., say YES if there is such a cycle, NO otherwise). Prove that the Subset Sum problem is polynomial time reducible to the Zero Weight Cycle Problem.

Given W, S

Construct V

$V := \{v_0, v_1, \ldots, v_n\}$

Add edges $e_{i1} = (v_{i-1}, v_i)$ with weight $= a_i$ Time: $O(n)$

$e_{i2} = (v_{i-1}, v_i)$ with weight $= 0$

Return edge

$e_R = (v_n, v_0)$ with weight $= -W$

($\Rightarrow$) If S has a subset, G has zeroweight cycle

let $\exists\, T \subseteq S$, $SUM(T) = W$

We construct a cycle from v_0 as

(i) $a_i \in T$: traverse inclusion edge (e_{i1})

(ii) $a_i \notin T$: " exclusion edge (e_{i2})

When we reach v_n (weight $= W$)

To complete the cycle, we traverse back to v_0 $(W = -W)$

Total cycle weight $= W + (-W) = 0$: zero weight cycle

($\Leftarrow$) If G has zero-weight cycle, subset sum $= W$

In G, edges as $v_{i-1} \to v$ except $v_n \to v_0$

Remove $v_n \to v_0$

It becomes a directed acyclic graph

While traversing $v_{i-1} \to v_i$ we have two choices — inc edge / exc edge

If we go through include edge, add a_i in T

At end go through $-W$

$\sum a_i - W = 0 \Rightarrow \sum a_i = W$

(valid subset of S with sum $= W$)

Hence Proved ∎

Que. Independent Set & Multiple Interval Scheduling

Construction:

let the Graph in IS be

$$G = (V, E)$$

$$S \subseteq G \quad s.t. \quad |S| = k$$

let edges for MIS problem be $e_1, e_2, \ldots, e_m$ $s.t.$ $m = |E|$

e_j corresponds to $[j, j+0.5]$

$\forall v_i \in V$, create job J_i

J_i requires time $[j, j+0.5]$ iff e_j incident on v_i

$$\text{Interval}(J_i) = \{ [j, j+0.5] \mid e_j \text{ incident on } v_i \}$$

$K' = k$

($\Rightarrow$) $S \subseteq V \mid |S| \geq K$

Jobs $J := \{ J_i \mid v_i \in S \}$

since $u, v \in V$; they have no common edge as e_j

J_u and J_v will not share common interval thus

Thus we schedule $\geq K$ jobs

($\Leftarrow$) $J_{\text{selected}} := \{ J_i \}$

Map them back to S

let S ~ not independent

for $u, v \in S$, $\exists e_j = (u, v) \in E$

J_u time $= [j, j+0.5]$

J_v time $= [j, j+0.5]$

Both require same interval, contradicts the fact J are non overlapping. S has independent vertices !

Hence Proved ■